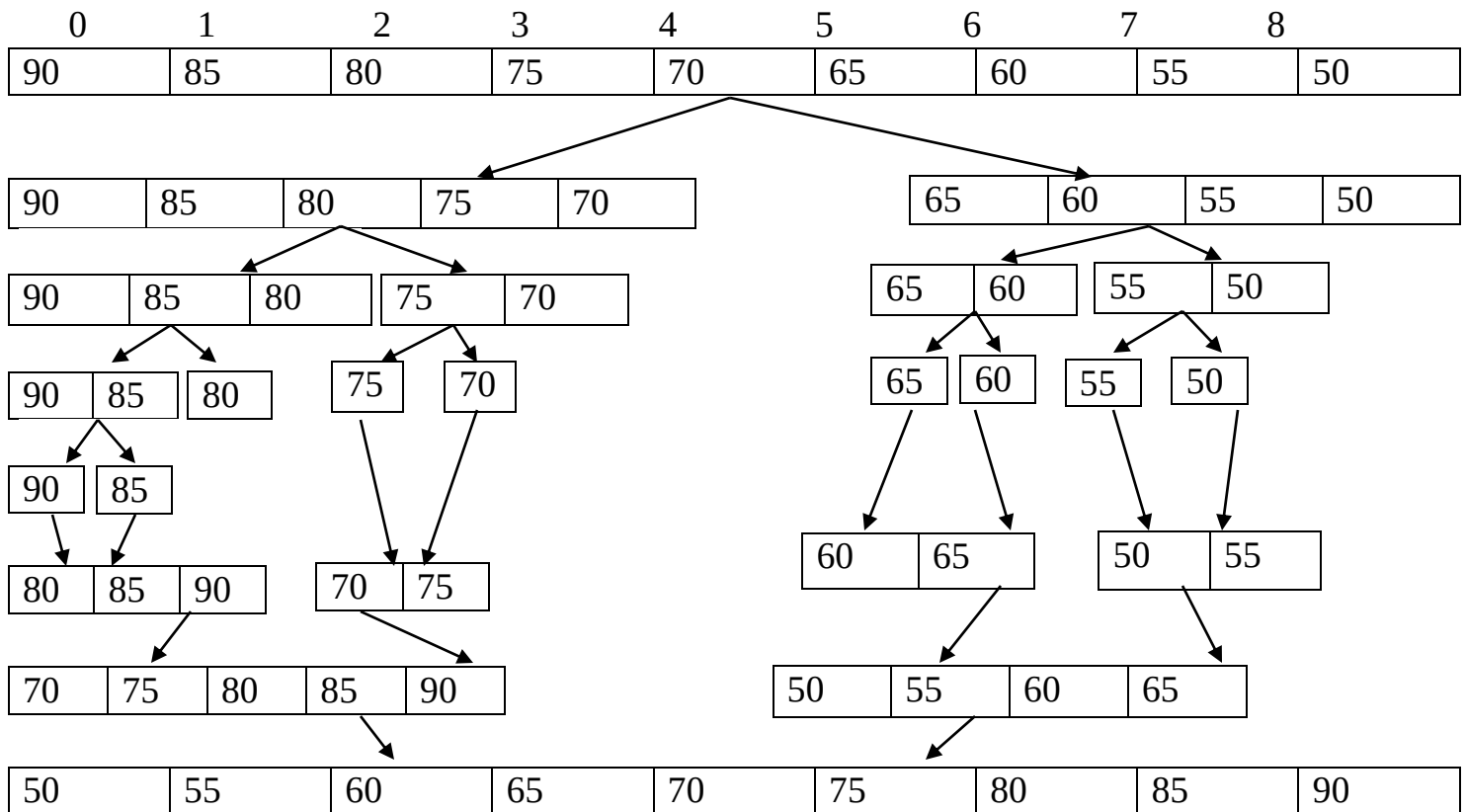


MERGE SORT ALGORITHM:-

Merge sort algorithm Conceptually, a merge sort works as follows:

1. Divide the unsorted list into n sub lists, each containing one element (a list of one element is considered sorted).
2. Repeatedly merge sub lists to produce new sorted sub lists until there is only one sub list remaining. This will be the sorted list.

Complexity of merge sort algorithm is **$O(n \log n)$** .



Algorithm:-

0 1 2 3 4 5 6 7 8

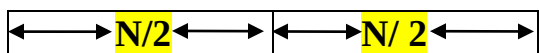
90	85	80	75	70	65	60	55	50
L				M			H	

Algorithm(Mergesort (L, H)

```
{
if ( L < H)
{
Mid = ( L + H)/2
mergesort(L,mid);
Mergesort(mid+1,H);
Merge(L,mid,H)
}
}
```

Merge Sort Time Complexity Analysis

Suppose **N** Size Array



$T(n) = T(n/2) + T(n/2) + n$
$T(n) = 2T(n/2) + n$ ----- eq(i)
substitute n/2 in place of n in eq (i)
$T(n/2) = 2T(n/4) + n/2$ -----eq(ii)
substitute eq (ii) in eq (i)
$T(n) = 2\{2T(n/4) + n/2\} + n$
$T(n) = 2^2T(n/2^2) + 2n$ -----eq(iii)
substitute n/4 in place of n in eq----(i)
$\therefore T(n/4) = 2T(n/8) + n/4$ -----eq(iv)
substitute eq(iv) back in eq(iii)
$T(n) = 2^2\{2T(n/8) + n/4\} + 2n$
$T(n) = 2^3T(n/2^3) + 3n$
So by finding pattern
$T(n) = 2^i T(n/2^i) + i n$ -----eq(iv)

$T(1) = 1$
let $n/2^i = 1$
$n = 2^i$
apply log on both sides.
$\log_2 n = \log_2 2^i$
$i \log_2 2 = \log_2 n$
$\log_2 n = i$
put in eq (iv)
$2^{\log_2 n} T(n/2^{\log_2 n}) + n \log_2 n$
$\therefore 2^{\log_2 n} = N$; Proved given below
$NT(N/N) + n \log_2 n$
$T(n) = NT(1) + n \log_2 n$
$T(n) = n \log n$ Proved

Prove that :- $2^{\log_2 N} = N$

Solution :-

Let $y = 2^{\log_2 N}$

taking $\log_2$ on both sides

$$\log_2 y = \log_2 2^{\log_2 N}$$

$$\log_2 y = 1 \cdot \log_2 N \quad (\because \log_2 2 = 1 \text{ or } \log_{10} 10 = 1)$$

$$\cancel{\log_2} y = \cancel{\log_2} N$$

$$y = N$$

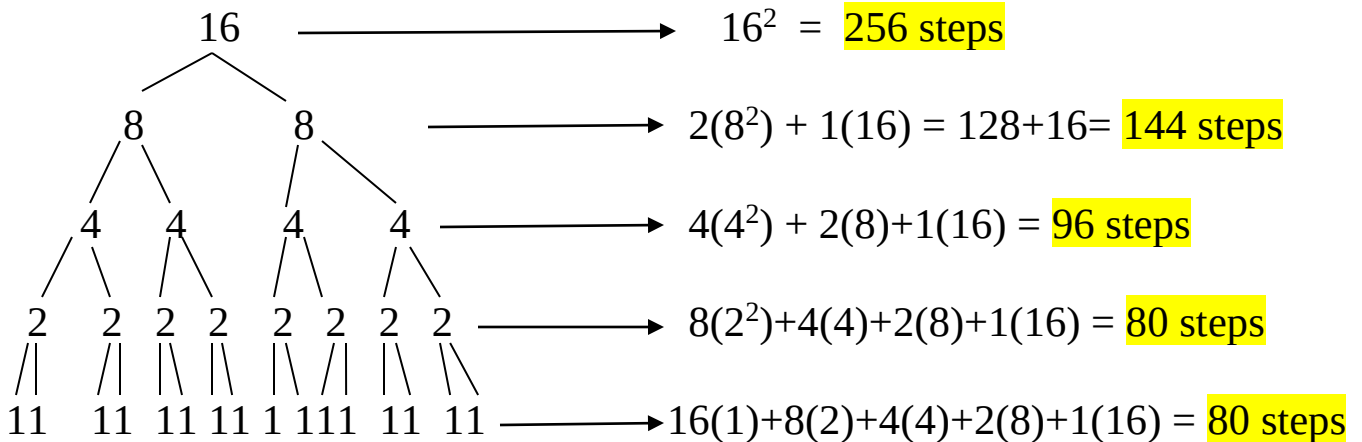
$$2^{\log_2 N} = N \text{ Proved}$$

Example :-

Assume you are having an algorithm for sorting an array known as bubble sort having the complexity $O(N^2)$. Try to improve the efficiency of sorting algorithm, that is reducing its complexity, and find its determinant complexity. (Assume $N=16$).

Solution :-

$$N=16 \longrightarrow O(N^2) = 16^2 \longrightarrow 256 \text{ steps}$$



Conclusion :-

Size of Array	Minimum steps Required.
16	80 ($16\log_2 16$)
32	192 ($32\log_2 32$)
64	448 ($64\log_2 64$)
N	$N\log_2 N$

64
 $32 \quad 32$
 $192 + 192 + 64 = 448$